

PLA protocol with Optional Concurrent IF

Updated by Brook Barajas 10/18/2017

****NOTE:** Make up wash buffers beforehand (from purchased packets or using recipes at the end of the protocol) and store in the fridge.

****NOTE:** Master mixes should be used for PLA steps whenever possible.

****NOTE:** You will need humidity chambers for this protocol – I have made them out of empty large 20µL tip boxes with the tip holder over the top, and lids from 1 mL tip boxes with the underside divider cut out. I then fill the bottoms with water and place in a 37°C incubator before starting the PLA section of the protocol. Anything that holds ½ inch of water, can be sealed, keeps the water away from the plate, and holds a 6-well plate can be used.

Also, you will need a dedicated 37°C incubator, or at least one with plenty of space that others aren't going into and out of during your PLA.

PART 1: PREPARE COATED CHAMBERS

****NOTE:** This is useful if you are using cells that tend to pile up, such as 293Ts. Piling and clumping make imaging and quantitation difficult.

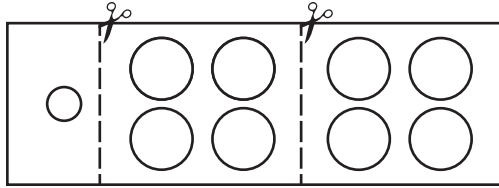
Day 1: Prepare poly-lysine coated chambers in 6-well plates.

Before starting, obtain:

- 6-well Falcon plates
- Sterile Poly-L-lysine
- Sterile 18x18mm coverslips (Zeiss, No. 1.5H 170±5µM thickness, item # 474030-9000-000)
- Sterile CultureWell multiwell 8-chambered coverslips (Grace Bio-labs, item # 103380)
- Sterilized forceps
- Sterilized scissors
- 200 µL Pipetman and sterile tips

***Unless specified, all of the following steps should be done in the hood.**

1. Label your sterile coverslips with a letter in the center (representing the experimental condition for that well) as well as a small dot in the upper left corner to orient you during mounting.
2. Peel a silicone chamber off the coverslip it comes on, and cut it between the two sets of 4 chambers and between the chambers and the topmost single hole:



3. Wipe down coverslips to remove all traces of dust. This can be done with your fingers if you are wearing new gloves that have been sprayed with 70% ethanol.
4. Place a silicone chamber on each coverslip, being careful to center them so that well edges are not too close to the edges of the coverslips. Press to seal thoroughly.
5. Trim excess silicone from the edges, orienting the coverslips with the glass facing you for easier visualization.
6. Place newly chambered coverslips into the bottom of 6-well plate wells, silicone side up.
7. Draw up a few hundred μL s of Poly-L-Lysine into a 200 μL Pipetman, and add enough to each silicone chamber to fill completely.
8. Once all chambers in a plate have been filled, place lids on plates and incubate at 37°C for 1 hour in a TC incubator.
9. In the hood, aspirate off the Poly-lysine, tilt plate and aspirate any final drops, replace the lid and allow to dry overnight, or for several days, in the TC hood.

PART 2: SEEDING AND TRANSFECTION

NOTES:

- Begin with cell plating 9 hours or more before transfection, so that cells have at least 7 hours to settle and adhere.
- With 293Ts, always plan to transfect within 18 hours of plating.
- Always harvest within 36-38 hours of transfection for Cos1 cells, 15-17 hours for 293Ts.
- VLP production is most efficient in cells between 5 and 30 passages. If your cells are at passage 28 or higher, you may need to thaw another aliquot before proceeding.
- **IMPORTANT:** If you will be seeding cells for parallel Western blots, designate separate plates for these – you will need to harvest them and their PLA counterparts at different times and under different conditions.

PART 2: PLATING CELLS

Day 2: plate cells into pre-prepared plates, then transfect with desired constructs.

Before starting cell plating:

- Make up and warm media.
- Warm PBS.
- Thaw an aliquot of 0.25% trypsin (don't leave in the water bath once it is warm, as this will decrease the efficiency). After use, label this with your initials and store in the TC fridge.

- If you have skipped the coating section, prepare wells with coverslips for the pLA. Write the condition identifier (A,B,C... or 1,2,3...) in the center of each coverslip and place a dot in the upper left corner. If re-using washed silicone wells, sterilize them, dry them, and center one on top of each coverslip, pressing them down with clean gloved fingers to ensure that they are adhered and checking for any bubbles. Next, place the coverslip with the glass-side down in a well, ensuring that the silicone insert is not touching the sides of the well. If not using the silicone wells, place a coverslip letter-side down on each slide.

Trypsinization:

1. Spray hood and all media/trypsin with 70% EtOH, wipe down with Kimwipes.
2. Remove 10 mL plates from the incubator and aspirate media.
3. Add 5 mL warmed PBS, tilt plate to wash, aspirate PBS.
4. Add 1 mL warmed trypsin, tilt plate to distribute.
5. Place plates in the incubator with the bottom of each plate in direct contact with the shelf to increase trypsinization efficiency.
6. Let sit 3-4 minutes (earlier passages are more adherent); if necessary, check under to microscope to ensure cells are lifting up off of the plate.
7. Quench trypsin by adding 5 mL 10% FBS DMEM + pen/strep to each plate, tilt the plate and wash with the same 5 mL of media several times to collect all cells.
8. Pipette quenched cells and media into a Falcon tube.
9. Tilt empty plates onto their sides, then aspirate the remaining media to prevent transfer of liquid to the biohazard waste bin.

Counting:

1. Once all cells are combined in a Falcon tube, pipette up and down to mix thoroughly, then remove a 50 μ L aliquot.
2. In a clean well of a 96 well plate, combine 40 μ L of cells with 40 μ L of trypan blue. Mix thoroughly.
3. Add 10 μ L of this 1:1 mix to each channel of a hemocytometer.
4. Count all four 4x4 corners of the hemocytometer grid, plus the center 16x16 square, for both sides of the hemocytometer.
5. Take the final number of cells from both wells combined, divide by 5, then multiply by 10,000 to get cell count per mL.

Plating:

****NOTE:** If seeding cells for parallel Western blots, use separate 6-well plates.

1. Before plating cells, add media to each plate so that the final volume including added cells will be 3.5 mL. **Ensure your chambered coverslips are completely submerged and have no large bubbles underneath.**
2. Thoroughly mix cells to resuspend, then add 1.5-3 million 293T cells, 600,000 Cos1 cells, or 500,000 HeLa cells per well into your 6-well plate. Numbers of 293T, HeLa and Cos1 cells are increased to promote better seeding of the 4-chamber silicone inserts. If using only coverslips without inserts, seed 1.3 million 293T, 300,000 Cos1 or 250,000 HeLa cells. Some constructs may cause more cell death – if you know that this is the case with any of the constructs you are planning to transfect, plate more cells in anticipation of cell death.

3. Gently slide plates back and forth and side to side across the incubator surface to distribute cells evenly.
4. Stack plates in pairs (no higher than two plates per stack, as tall stacks can lead to vibration effects and spillage) and place in the incubator.

PART 3: TRANSFECTION

Before starting cell transfection:

- Warm serum/antibiotic-free DMEM ("SFM" or serum-free medium).
- Warm transfection medium (10% FBS DMEM without antibiotics).
- Thaw the appropriate constructs by placing in a 50°C thermomixer with rotation for at least 15 minutes to fully resuspend.
- Retrieve the appropriate volume of 1 mg/mL PEI from the Chesebro tower in the -80 and thaw at 50° C in the TC room thermomixer.
- Prepare pipetmen by wiping their barrels with EtOH and placing them in the hood.

Transfection:

NOTE: At least 7 hours after plating, cells are ready for transfection. You can start to set up your transfection mixes as early as 5 hours after plating. HOWEVER, remember that you must harvest the lysates between 36-38 hours after transfection for Cos-1 cells, 15-17 hours after for 293Ts.

1. For each construct, label a sterile Eppendorf tube and add 250µL SFM. If transfecting multiple wells with the same construct, make up a master mix, then add 260µL of this to each well.
2. Add the predetermined amount (1-10 µg) of plasmid DNA.
3. Let sit 5 minutes.
4. Add 10 µL thawed PEI from the stock tube; vortex vigorously and spin briefly to collect the droplets.
5. Let this mix stand at room temperature for 15 minutes.
6. Meanwhile, change the medium on your plates to 3 mL of transfection medium using sterile TC pipettes. Bring plates out 1 at a time and put each batch back into the incubator when done.
7. Add the DNA/PEI mix dropwise to the cells using a 1 mL Pipetman, distributing it around the plate.
8. Gently slide plates back and forth and side to side across the incubator surface to mix.
9. Spray the hood with Wescodyne to disinfect and wipe down, then spray with ethanol and wipe down again.
10. The next morning, replace the media with 3.5 mLs new transfection media.

PART 4: CELL FIXATION, PERMEABILIZATION, AND BLOCKING

Before starting cell fixation:

- Obtain cold and room temperature PBS for washes.
- Prepare room temperature 4% paraformaldehyde in PBS pH 7.4 (Make up 2mL per well, plus 10% extra for pipetting error, **right before starting this part of the protocol**). PFA will rapidly polymerize in aqueous solution.
- Make up permeablization solution (2mL per well, plus 10% extra for pipetting error): PBS + 0.5% saponin.
 - **NOTE:** some antibodies work better with 0.3% TritonX permeablization (including Jon's FIV p24 antibody)
- Prepare or retrieve PLA blocking solution – the Duolink version is stored at 4°C.
- Fill the bottom of your humidity chamber(s) with ½" of water and place in a 37°C incubator (see note in the beginning of the protocol).
- **Bring Wash Buffers A and B to room temperature in preparation for subsequent steps.**

1. Remove the PLA plates from the incubator and place on ice.
2. Aspirate the media from the plates and add 2mL ice-cold PBS per well.
3. Tilt plate 5x to rinse chambers.
4. Remove the PBS and repeat this wash.
5. Tilt each plate onto its side to collect the last of the PBS and aspirate this off.
6. Add 2mL 4% PFA to each well, ensuring that chambered coverslips are submerged.
7. Incubate for 15 minutes at room temperature.
8. Remove the PFA and rinse cells twice with room temperature PBS.
9. After the last wash, tilt each plate onto its side to collect the last of the PBS and aspirate this off.
10. Add 2mL permeablization solution to each well.
11. Incubate for 10 minutes at room temperature.
12. Remove permeablization solution.
13. Add 2mL PBS to each well and place on a rotator, speed 2.5-3.5, for 5 minutes.
14. Aspirate PBS and repeat this wash twice more.
15. After the last wash, aspirate PBS from the wells.
16. Use a Q-tip to carefully dry and remove cells from the top of the silicone insert around the wells to prevent well mixing. This must be done VERY gently so as not to peel up the wells from the coverslip, or press the chambered coverslip against the walls of the well. Anchor the chambered slide to the bottom of the well with forceps if necessary.
17. Fill each chamber with blocking solution.
18. Incubate the slides in the pre-heated humidity chamber for 30 min at 37°C.

PART 5: PRIMARY ANTIBODY INCUBATION

Before starting:

- Dilute your primary antibody (IF alone) or antibodies (PLA or PLA-IF) to suitable concentration in the Duolink antibody diluent. 1:250 is the recommended starting concentration. Make up 25µL antibody solution per sample the first time through the protocol, then scale down later if you end up with more than you need.
1. For one plate at a time, tilt the plate and quickly but carefully aspirate the blocking solution from the sample area.

2. Using a 200µL Pipetman, immediately add 1 drop (approximately 15-20µL) of the primary antibody solution to each chamber.
3. Place the plate back into the humidity chamber, then move on to the next plate and repeat.
4. Incubate at 37°C for 1 hour.
5. While samples are incubating, harvest your Western samples.

PART 6: PROBE INCUBATION

****NOTE:** For IFs alone (without PLA), add antibody diluent to wells.

Before starting:

- Dilute your two PLA probes 1:5 in antibody diluent. Make up 25µL of probe solution per sample. *E.g. for a 25 µL reaction add 5 µL of PLA probe MINUS stock, 5 µL of PLA probe PLUS stock to 15 µL of Antibody Diluent.*
1. For one plate at a time, tilt the plate and quickly but carefully aspirate the primary antibody from the sample area.
 2. Add 2 mLs 1x Wash Buffer A to each well, ensuring chambered coverslips are submerged.
 3. Repeat with remaining plates.
 4. Place plates on a gentle rocker and wash for 5 minutes.
 5. Repeat for a second wash.
 6. Again, for one plate at a time, tilt the plate and quickly but carefully aspirate the wash buffer from the wells and sample area.
 7. Using a 200µL Pipetman, immediately add 1 drop (approximately 20µL) of the PLA probe solution (or diluent only for IFs alone) to each chamber. Do not overfill, as you don't want the chambers mixing over the top of the silicone insert.
 8. Place the plate back into the humidity chamber, then move on to the next plate and repeat.
 9. Incubate for 1 hour at 37°C.

PART 7: LIGATION

****NOTE:** For IFs without PLA, add antibody diluent to wells after washes instead of ligation mix.

Before starting:

- Make up 25µLs of 1x ligation solution per chamber
 - Dilute the 5x Ligation solution stock 1:5 in high purity water and mix. *Wait to add the Ligase until immediately before addition to the samples. Take the addition of Ligase into account when calculating the amount of water added. E.g. for a 25 µL reaction add 5 µL of the 5x Ligation stock to 19.375 µL of high purity water.*
1. For one plate at a time, tilt the plate and quickly but carefully aspirate the primary

- antibody solution from each chamber.
2. Add 2 mLs 1x Wash Buffer A to each well ensuring chambered coverslips are submerged.
 3. Repeat with remaining plates.
 4. Place plates on a gentle rocker and wash for 5 minutes.
 5. Repeat for a second wash.
 6. Remove the Ligase from the freezer using a freezing block (-20 °C). Add Ligase to the Ligation solution from step a) at a 1:40 dilution and vortex. *E.g. for a 25 µL reaction add 0.625 µL of Ligase to 24.375 µL of prepared 1x Ligation solution.*
 7. Again, for one plate at a time, tilt the plate and quickly but carefully aspirate the wash buffer from the wells and sample area.
 8. Using a 200µL Pipetman, immediately add 1 drop (approximately 20µL) of the Ligation-Ligase solution to each chamber. For IFs, instead add antibody diluent.
 9. Place the plate back into the humidity chamber, then move on to the next plate and repeat.
 10. Incubate for 30 min at 37°C.
 11. Add 2 mL 1x Wash Buffer A to each well ensuring chambered coverslips are submerged.
 12. Repeat with remaining plates.
 13. Place plates on a gentle rocker and wash for 2 minutes.
 14. Repeat for a second wash.

PART 8: AMPLIFICATION

****NOTE:** For IFs alone, add secondary antibody here.

****NOTE:** While the incubation step in this section is underway, you can label your slides for the mounting step.

Before starting:

- Dilute the Amplification stock 1:5 in high purity water and mix. *Wait to add the Polymerase until immediately before addition to the sample. Take the addition of Polymerase into account when calculating the amount of water added. E.g. for a 25 µL reaction add 5 µL of the 5x Amplification stock to 19.69 µL of high purity water.*
- Remove the Polymerase from the freezer using a freezing block (-20 °C).
- Add Polymerase to the Amplification solution at a 1:80 dilution and mix by pipette. *E.g. for a 25 µL reaction add 0.31 µL of Polymerase to 24.69 µL of Amplification solution.*
- Make up 2mL 0.01x Wash Buffer B for each sample, plus an extra 10% volume for pipetting error.
- For IFs alone, make up secondary antibody in antibody diluent.

****NOTE:** This step is light sensitive; photobleaching of the fluorophore can occur with light exposure. Cover the lids of your plates in foil.

1. For one plate at a time, tilt the plate and quickly but carefully aspirate the wash buffer from the wells and sample area.
2. Using a 200µL Pipetman, immediately add 1 drop (approximately 20µL) of the Amplification-Polymerase solution to each chamber.
3. Place the plate back into the humidity chamber, cover with foil, then move on to the

- next plate and repeat.
4. Incubate for 100 min at 37°C.
 5. For one plate at a time, tilt the plate and quickly but carefully aspirate the Amplification-Polymerase solution from the sample area.

PART 9, OPTION 1: NO IF

1. Add 3 mL 1x Wash Buffer B to each well.
2. Repeat with remaining plates.
3. Cover plates with foil and place on a gentle rocker to wash for 10 minutes.
4. Repeat for a second wash.
5. For the 3rd wash, add 3 mL 0.01x Wash Buffer B, cover with foil and place on gentle rocker for 1 min.
6. Tilt the plates and quickly but carefully aspirate the wash buffer from the well and sample area.
7. Proceed to mounting

PART 9, OPTION 2: CONCURRENT IF

****NOTE:** Concurrent IF referred to here is against one of the two target proteins in the PLA. The fluorophore of the secondary IF antibody used must be a different color than your PLA, and should be well-separated in both emission and excitation from your PLA fluorophore. We use the red PLA kit with Alexafluor 488 secondaries for IF, or the green PLA kit with Alexafluor 594 secondaries.

****NOTE:** Here, antibody diluent without secondary should be used on samples that will not get a concurrent IF.

Before starting:

- Using the guidelines above, make up stocks of secondary antibody (we often use 1:1000 as a starting point) against your PLA target of interest in antibody diluent. Again, start with about 25µL per sample.
1. Add 3 mL 1x Wash Buffer B to each well.
 2. Repeat with remaining plates.
 3. Cover plates with foil and place plates on a gentle rocker to wash for 5 minutes.
 4. Repeat for a second wash.
 5. Fully remove wash buffer from wells, then from silicone around chambers, then from the chambers themselves with a P200 tip aspirator.
 6. Using a 200µL Pipetman, add secondary antibody solution to each concurrent IF chamber, and antibody diluent alone to all other chambers.
 7. Incubate for 15 minutes at room temperature, covered in aluminum foil
 8. For one plate at a time, tilt the plate and quickly but carefully aspirate the antibody solution or antibody diluent from the chamber.
 9. Add 3 mL 1x Wash Buffer B to each well.
 10. Repeat with remaining plates.
 11. Cover plates with foil and place plates on a rocker to wash for 10 minutes.
 12. Repeat for a second wash.

13. For the 3rd wash, add 3mL 0.01x Wash Buffer B, cover with foil and place on rocker for 1 min.
14. Proceed to mounting.

PART 9, OPTION 3: SEQUENTIAL IF

****NOTE:** Sequential IF referred to here is NOT against one of the two target proteins in the PLA. It is especially useful if you are doing non-immune control PLA on transfected cells and want to quantify your negative control spots in cells containing your transfected protein. As with concurrent IF, the fluorophore of the secondary IF antibody used must be a different color than your PLA, and should be well-separated in both emission and excitation from your PLA fluorophore. We use the red PLA kit with Alexafluor 488 secondaries for IF, or the green PLA kit with Alexafluor 594 secondaries.

****NOTE:** Here, antibody diluent without primary should be used on samples that will not get a sequential IF. For those chambers in the same well that will get a concurrent IF, incubate them in antibody diluent while the sequential IF samples are in primary, then proceed with the concurrent IF secondary incubation at the same time as you are doing the secondary incubation for the sequential samples.

****NOTE:** It is important to include a positive control PLA chamber with concurrent IF or fluorescent protein in the same well as your negative controls, to go through the same alternate wash and incubation steps (but with antibody diluent where appropriate). See previous note for how to proceed with these mixed wells.

Before starting:

- Using the guidelines above, make up stocks of primary antibody and secondary antibody (we often use 1:1000 as a starting point for secondary) against your third target of interest in antibody diluent. Again, start with about 25µL per sample.
1. Add 3 mL 1x Wash Buffer B to each well.
 2. Repeat with remaining plates.
 3. Cover plates with foil and place on a gentle rocker to wash for 5 minutes.
 4. Repeat for a second wash.
 5. Using a 200µL Pipetman, add primary antibody solution to each sequential IF chamber, and antibody diluent alone to all other chambers.
 6. Incubate for 30 minutes at 37°C. (This incubation time can be increased if the signal is found to be too weak).
 7. For one plate at a time, remove primary antibody or antibody diluent from the chambers.
 8. Quickly add 2 mLs Wash Buffer A to each well.
 9. Repeat with remaining plates.
 10. Cover plates with foil and place on a gentle rocker to wash for 5 minutes.
 11. Repeat this wash.
 12. Fully remove wash buffer from wells, then from silicone around chambers, then from the chambers themselves with a P200 tipped aspirator.
 13. Using a 200µL Pipetman, add secondary antibody solution to each concurrent IF chamber, and antibody diluent alone to all other chambers.
 14. Incubate for 15 minutes at room temperature, covered in aluminum foil

15. For one plate at a time, tilt the plate and quickly but carefully aspirate the secondary antibody solution from the sample area.
16. Add 3 mL 1x Wash Buffer B to each well.
17. Repeat with remaining plates.
18. Cover plates with foil and place plates on a rocker to wash for 10 minutes.
19. Repeat for a second wash.
20. For the 3rd wash, add 3mL 0.01x Wash Buffer B, cover with foil and place on rocker for 1 min.
21. Proceed to mounting.

PART 10: MOUNTING

Before starting:

- Gather materials:
 - 1 bottle of clear quick-dry nail polish.
 - 1mL syringe with a 20-gauge needle attached.
 - PLA mounting medium plus DAPI.
 - Forceps
 - 1 large light-proof box lined with Kimwipes
 - 1 slide box
 - Your labeled glass microscope slides (we use Fisher Scientific Fisherbrand Superfrost/Plus 25x75x1mm slides, catalog # 12-550-15)
1. Gently remove the chambered coverslip from your first well, replacing the foil cover on the plate afterwards.
 2. Peel away the silicone chamber and set aside.
 3. Carefully aspirate off all wash buffer.
 4. Place cell side up on the Kimwipes in the prepared light-proof box, and close the lid.
 5. Repeat for all coverslips.
 6. Let dry 10-15 min. Ensure that your coverslips are completely dry.
 7. Prepare a slide, placing labeled-side up and left on the bench.
 8. Draw up several hundred microliters of mounting medium into your syringe.
 9. Using forceps, gently lift a coverslip from the box, then hold gently by the sides, cell side up.
 10. Re-cover the box until ready to mount the next sample.
 11. Holding the coverslip sample-side up, place a single drop of mounting medium into the center, then quickly invert onto the slide and gently press down to remove bubbles and spread the medium.
 12. Aspirate any excess mounting medium from around the edges of the coverslip.
 13. Use nail polish to paint around the edges of your coverslip, sealing it to the slide.
 14. Place face up in the slide box to keep it in the dark, and proceed to the next coverslip.
 15. Let slides “cure” for 24 hours at RT, then place at -20°C until ready to image.
 16. Wait for approximately 24 hours before analyzing in a fluorescence or confocal microscope, using at least a 20x objective to visualize spots.
 17. After imaging, store the slides at -20°C in the dark. *Duolink In Situ Mounting Media with DAPI is aqueous and does not solidify.*

BUFFERS:

To make your own Duolink In Situ Wash Buffer A:

Dissolve 8.8 g NaCl, 1.2 g Tris base and 0.5 ml Tween 20 in 800 ml high purity water. Adjust pH to 7.4 using HCl. Add high purity water to 1000 ml (final concentrations 0.01 M Tris, 0.15 M NaCl and 0.05% Tween 20). Filter the solution through a 0.22 μm filter and store at +4 °C. Bring the solutions to room temperature before use.

To make your own Duolink In Situ Wash Buffer B:

Dissolve 5.84 g NaCl, 4.24 g Tris base and 26.0 g Tris-HCl in 500 ml high purity water. Adjust pH to 7.5 using HCl. Add high purity water to 1000 ml (final concentrations 0.2 M Tris and 0.1 M NaCl). Filter the solution through a 0.22 μm filter and store at +4 °C. Bring the solutions to room temperature before use.